

Measuring Impact: New Statistical Methods for Assessment Accuracy and Transparency

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Cook County, Illinois, USA



Data and Modeling

Outlier sales exist. How should we identify and exclude them?

Data errors exist. How do we find them?

Past ratio stats have room for improvement. Does a non-linear model help?

Transparency problems

We've improved the data and modeling. **How can we earn trust after years of** public mistrust and documented assessment inequities?

Today's Agenda



Outlier sales

- Analytic approaches
- Technical implementation

Data errors

- Our process to automatically tag and fix data errors

A new model

- Linear vs. tree-based model: does it matter?

Explainability

- A new sale comps algorithm

Cook County

Cook County, Illinois, USA:

- Largest market-based assessment jurisdiction in the United States
- 1.9M parcels (2024)
- Population: 5.1M (2022)

Tax Facts:

- Over 130 municipalities
- 941 taxing agencies for roughly 800 units of government
- Levy-based; the rate floats



Cook County Property Taxes: By the Numbers



Property values and tax bills vary throughout Cook County.

- Chicago \$220k home → \$4k bill (1.8%)
- Winnetka \$1.08M home → \$24k bill (2.2%)
- Dixmoor \$66k home → \$2.2k bill (3.3%)

With 12 or more taxing agencies on each homeowner's bill, tax rates that change every year, and some of the US's highest effective tax rates, **people frequently demand change.**

OPINION

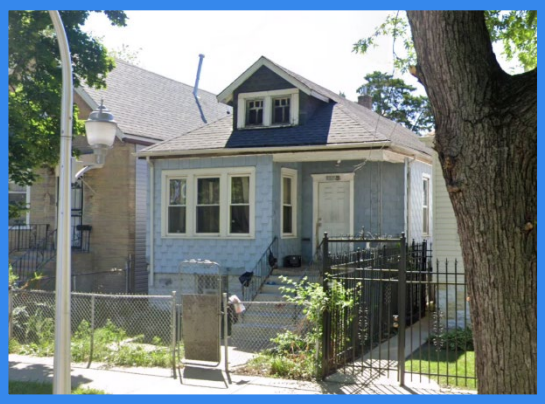
“ How Lower-Income Americans Get Cheated on Property Taxes

Many homeowners are paying a total of billions of dollars extra
because of inequities in assessing property values.

”

April 3, 2021

Chicago



\$75K estimate

\$100K sale



\$800K estimate

\$1M sale

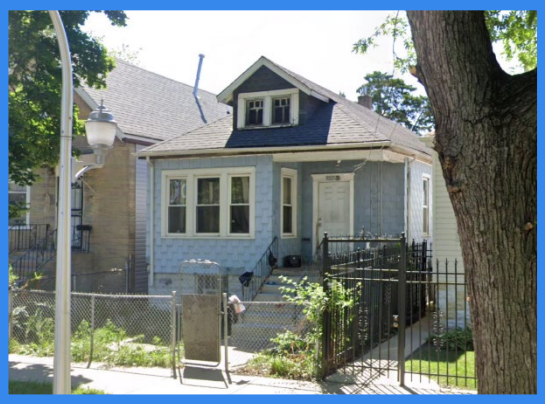
Bottom 10%

Top 10%

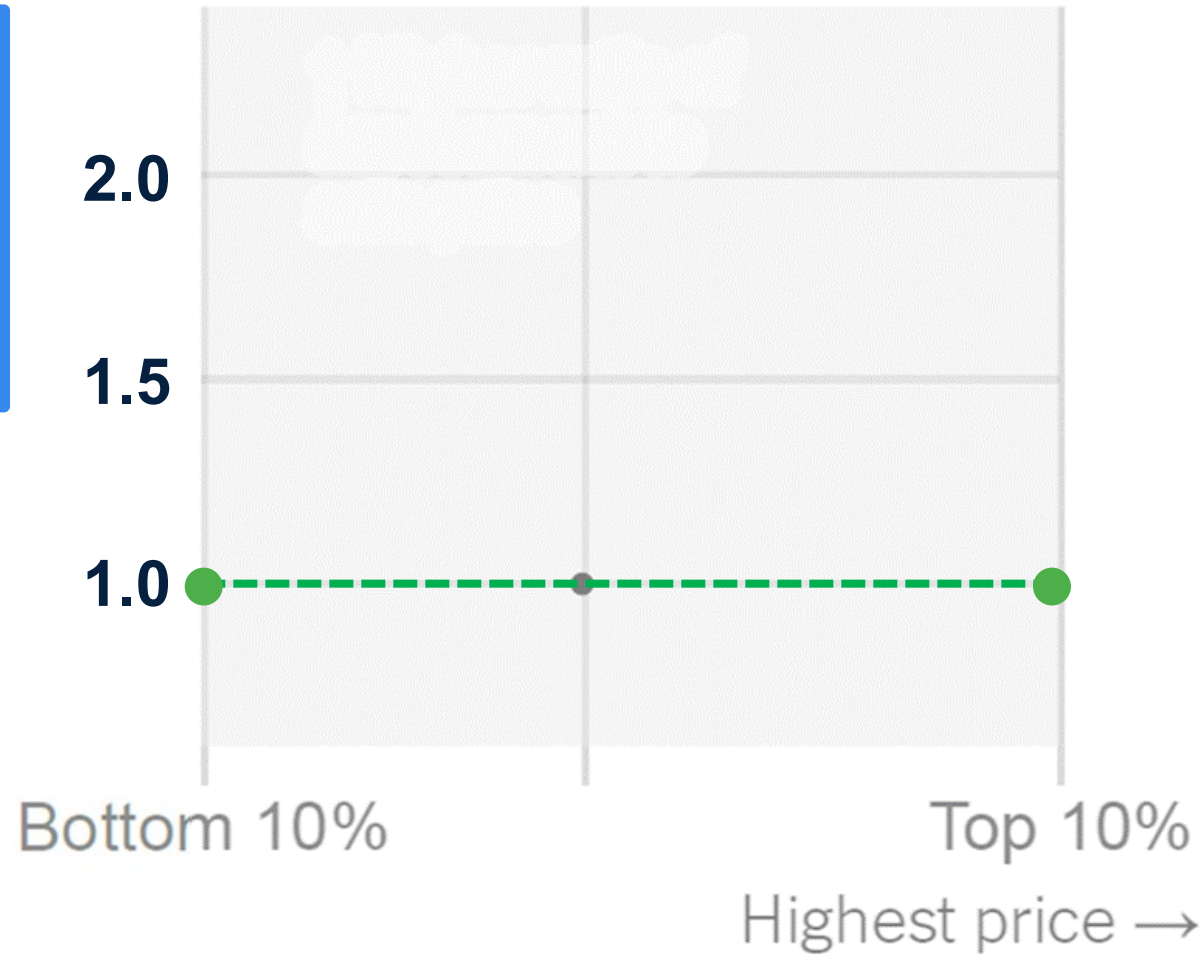
Highest price →

Source: **Chris Berry/NYT**, using assessments from prior years

Chicago



\$100K estimate
• **\$100K** sale
= **1.0 ratio**



\$1M estimate
• **\$1M** sale
= **1.0 ratio**

Source: **Chris Berry/NYT**, using assessments from prior years

Chicago



\$150K estimate
÷ **\$100K** sale
= **1.5 ratio**



\$800K estimate
÷ **\$1M** sale
= **0.8 ratio**

Source: **Chris Berry/NYT**, using assessments from prior years

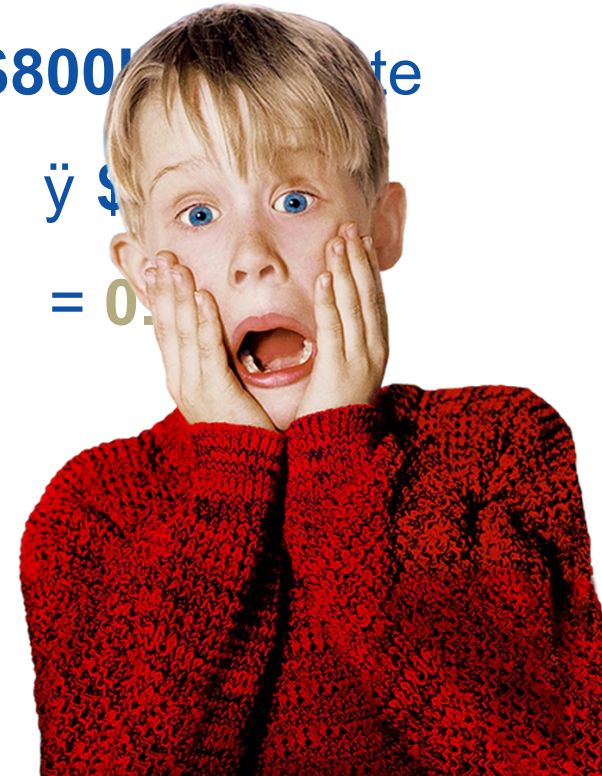
Chicago



\$150K estimate
• **\$100K** sale
= **1.5 ratio**



\$800K estimate
• **\$800K** sale
= **1.0 ratio**



Source: **Chris Berry/NYT**, using assessments from prior years

Sales
&
characteristics



The Model



High-quality
assessments

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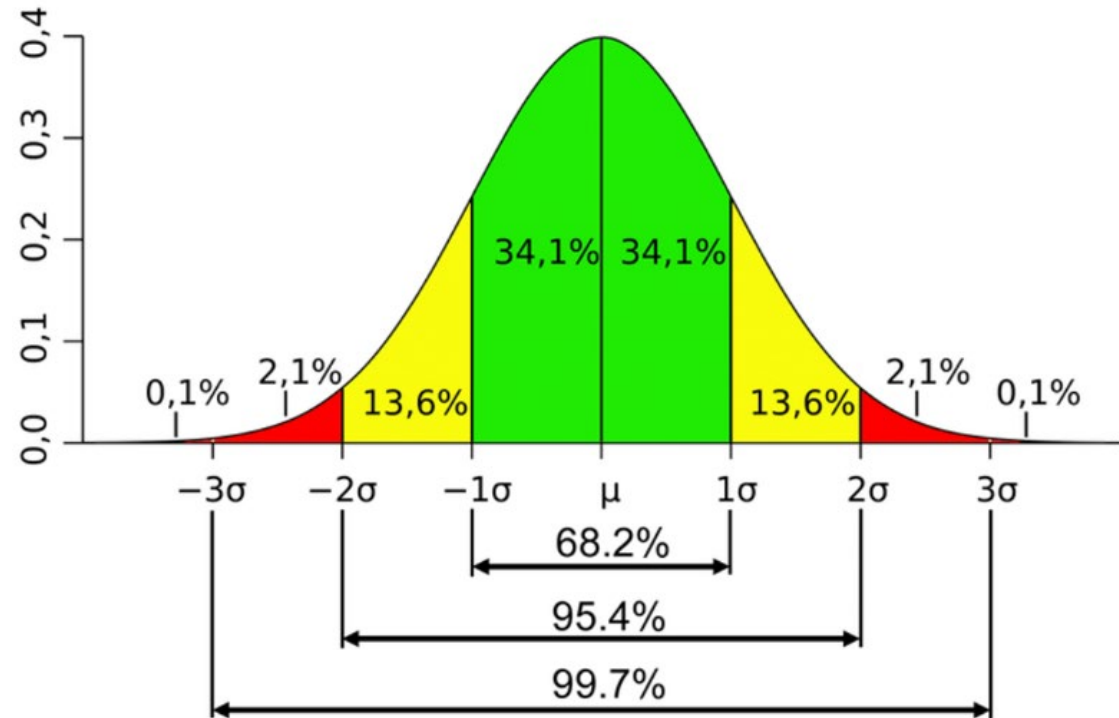
Outlier Sales

 @ccao-data/model-sales-val

► *Problem:* some sales are **outliers** (e.g, a home that sells for 50% higher than similar homes.)

 *Why it matters:* Garbage in, garbage out.

 *Solution:* build a pipeline to identify and exclude statistical outlier sales (~7% of sample).

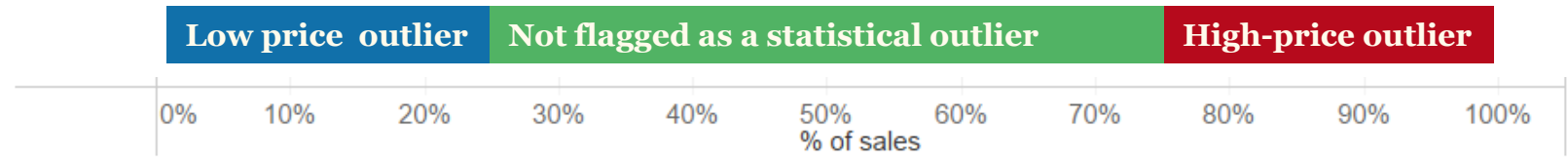


Statistical flags

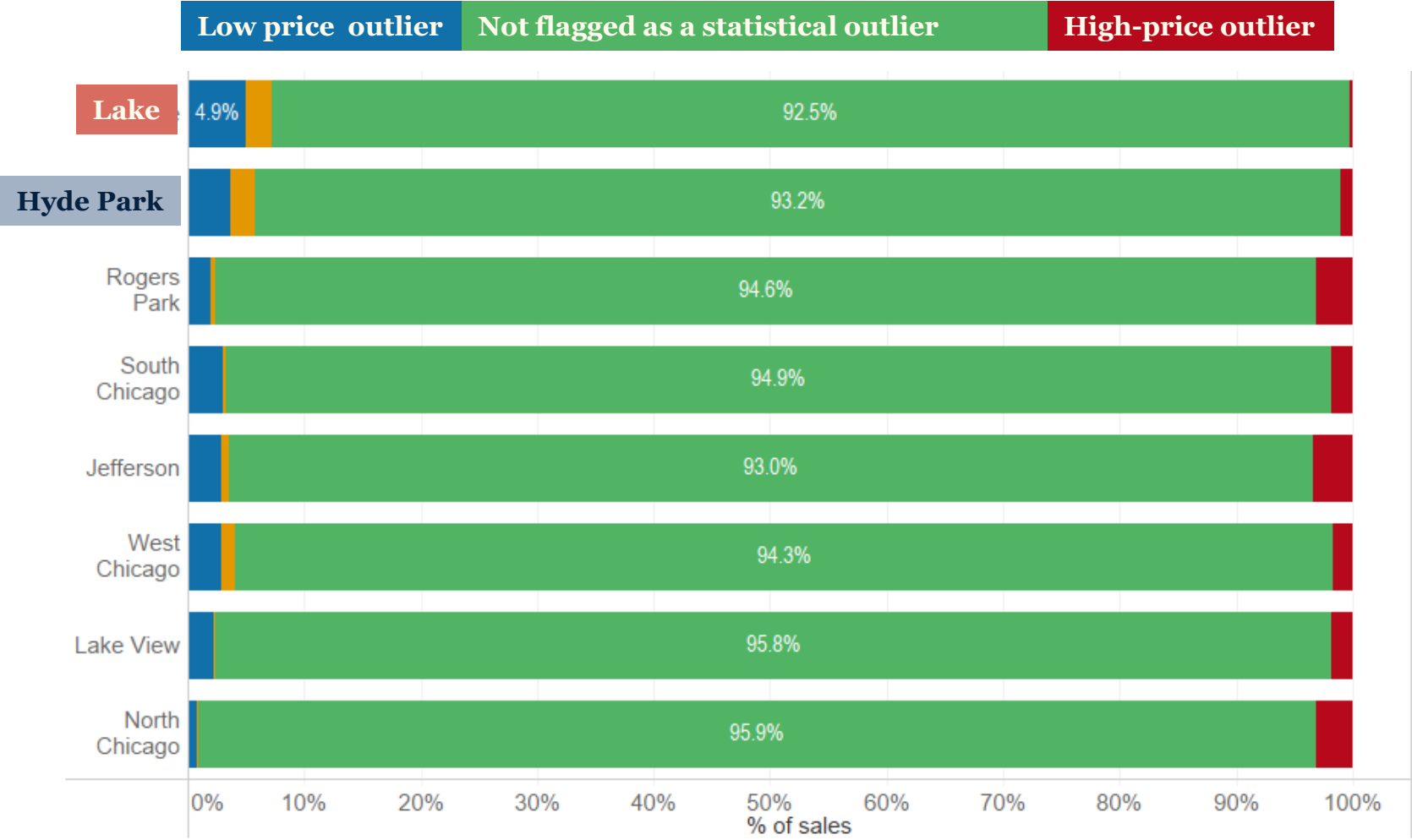
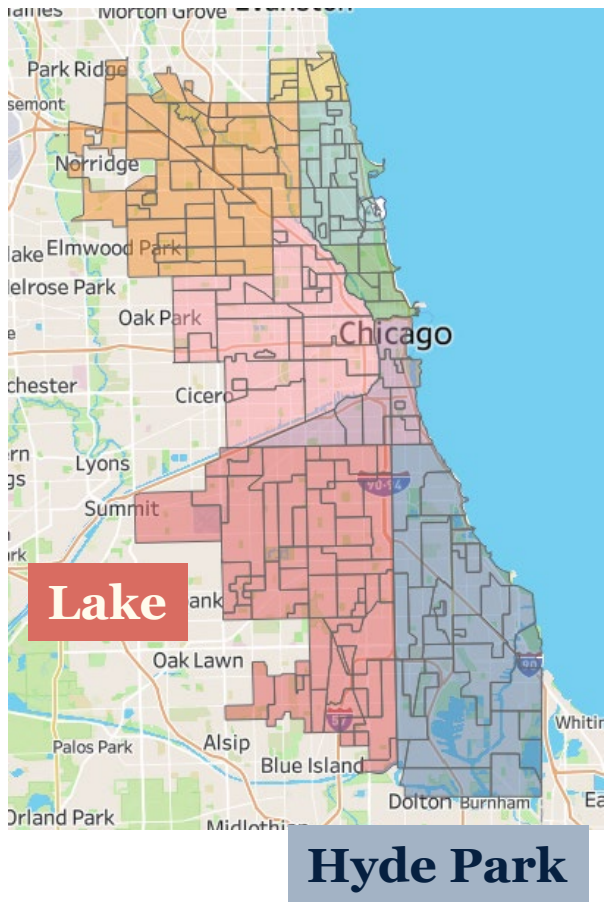
High Price	Special flags + high price outlier		
	High price outlier		
		High price (raw)	High price (+2 std)
		High price (per sqft)	High price per sq. ft. (+2 std)
Low Price	Special flags + low price outlier		
	Low price outlier		
		Low price (raw)	Low price (-2 std or under \$10k)
		Low price (per sqft)	Low price per sq. ft. (-2 std)

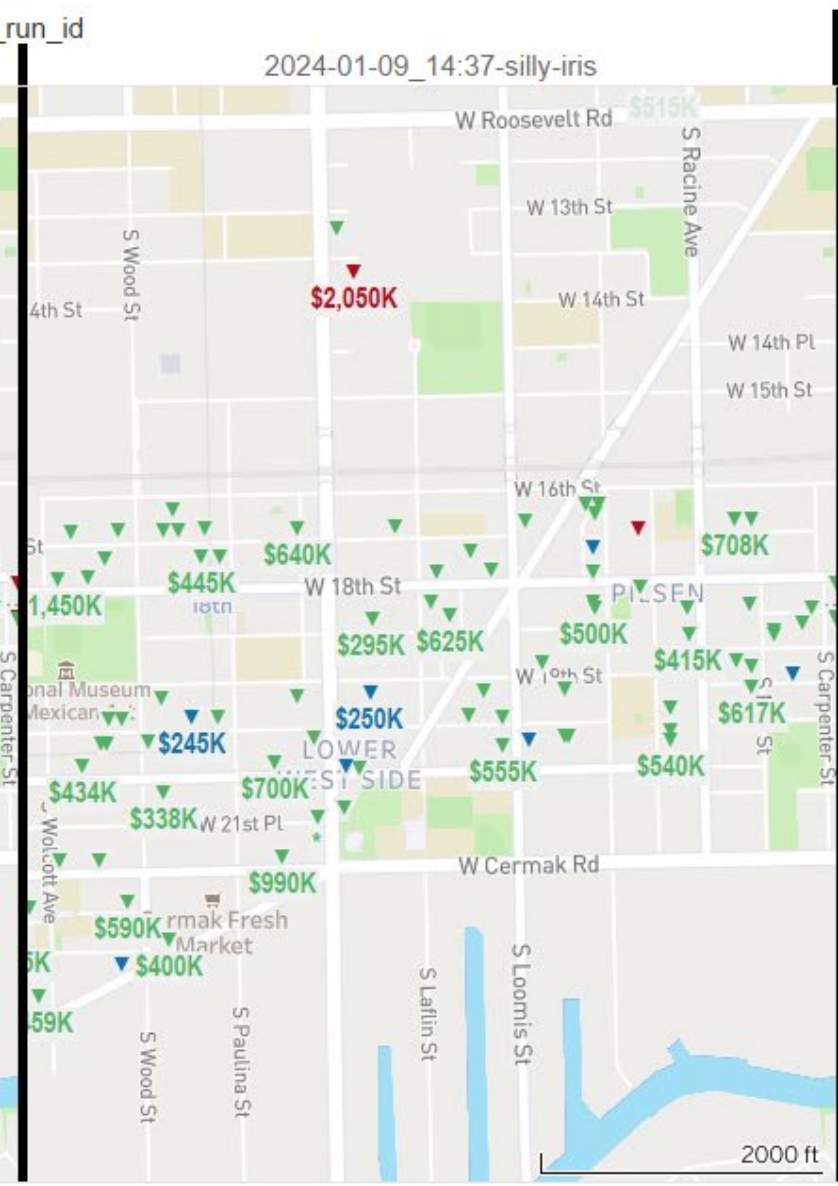
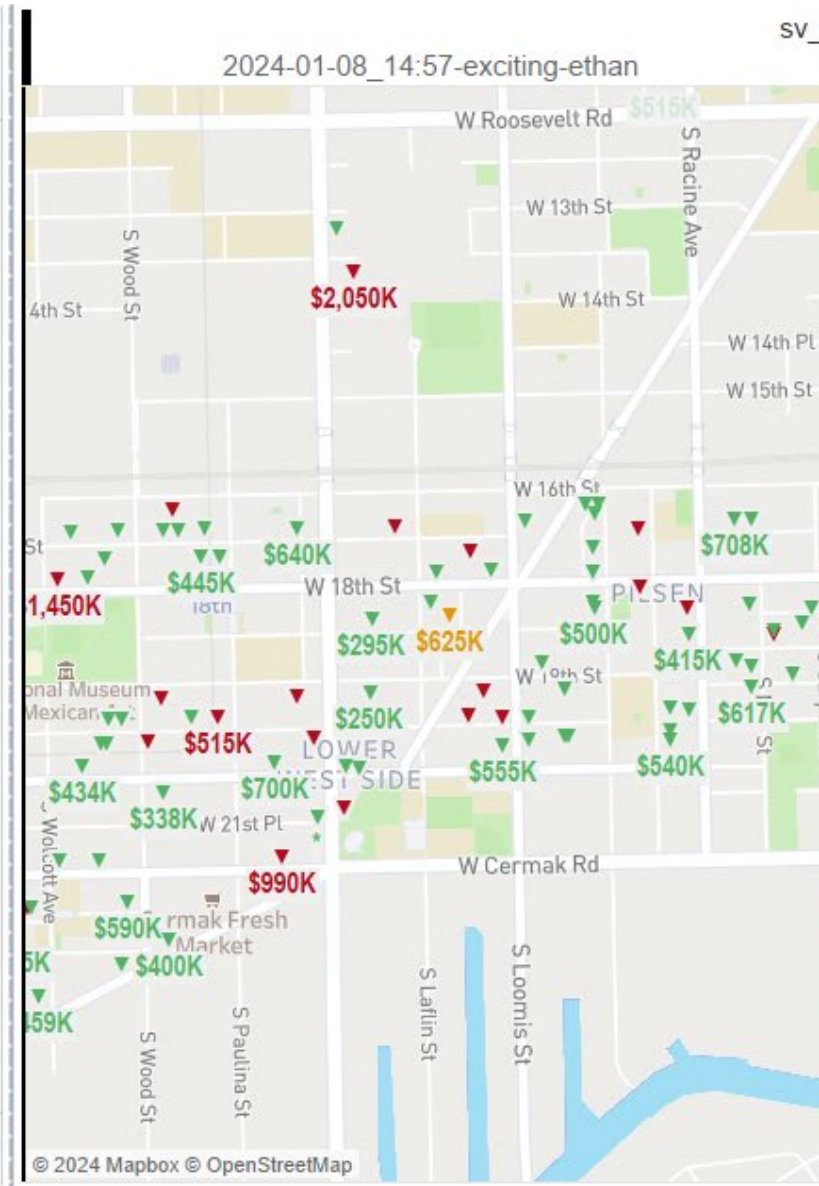
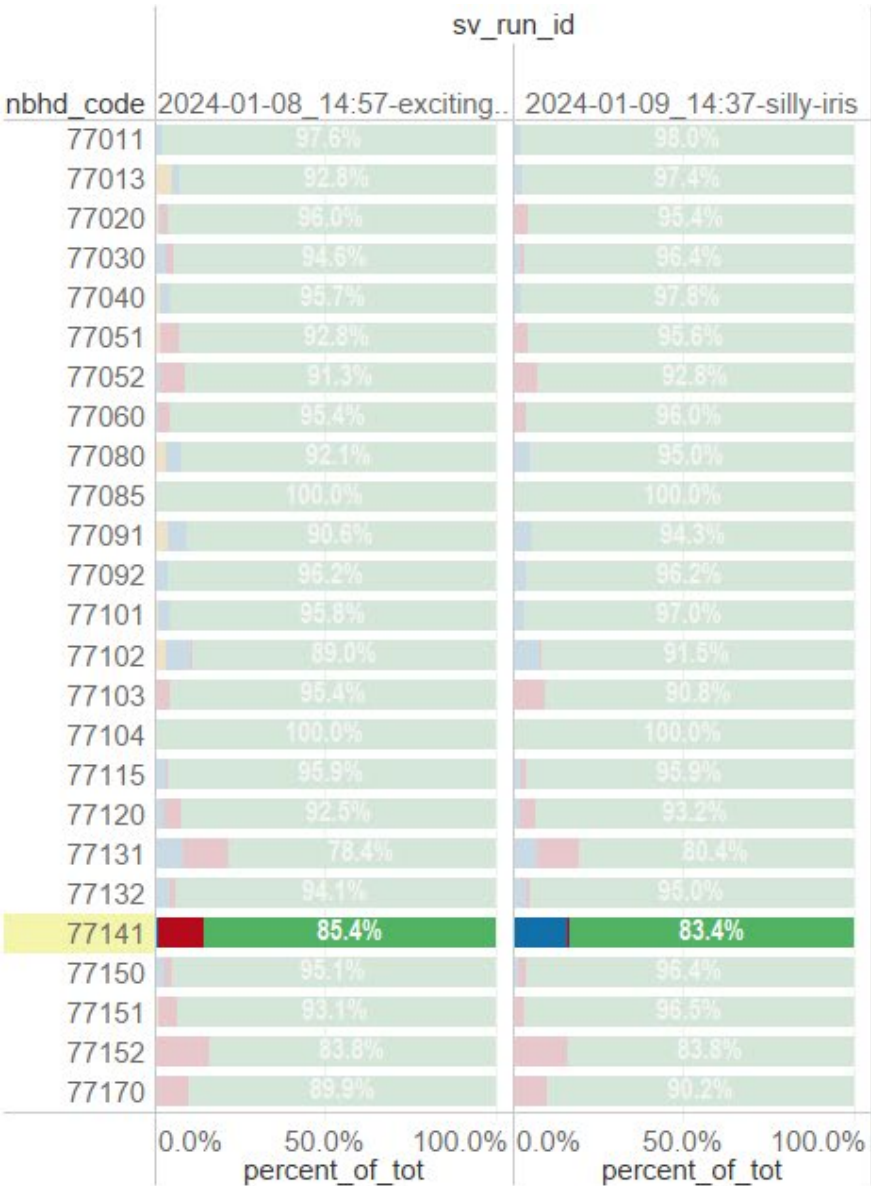
Statistical flags

High Price	Special flags + high price outlier	Home flip sale (high)	Short-term owner < 1 year & [1 high statistical outlier type]	Multiple flags
		Family sale (high)	Last name match & [1 high statistical outlier type]	
		Non-person sale (high)	Legal / corporate entity + [1 high statistical outlier type]	
	High price outlier	Anomaly algorithm (high)	Anomaly algorithm & [1 high price statistical outlier type]	
		High price (raw & sqft)	High price (+2 std) & high price per sq. ft. (+2 std)	
		High price swing	Large swing from mean (+2 std) & high price outlier (+2 std)	
		High price (raw)	High price (+2 std)	
		High price (per sqft)	High price per sq. ft. (+2 std)	
Low Price	Special flags + low price outlier	Home flip sale (low)	Short-term owner < 1 year & [1 low statistical outlier type]	Ownership "hiccups"
		Family sale (low)	Last name match & [1 low statistical outlier type]	
		Non-person sale (low)	Legal / corporate entity + [1 low statistical outlier type]	
	Low price outlier	Anomaly algorithm (low)	Anomaly algorithm & [1 low price statistical outlier type]	
		Low price (raw & sqft)	Large swing away from mean (-2 std) & low price outlier (-2 std)	
		Low price swing	Low price (-2 std) & low price per sq. ft. (-2 std)	
		Low price (raw)	Low price (-2 std or under \$10k)	
		Low price (per sqft)	Low price per sq. ft. (-2 std)	

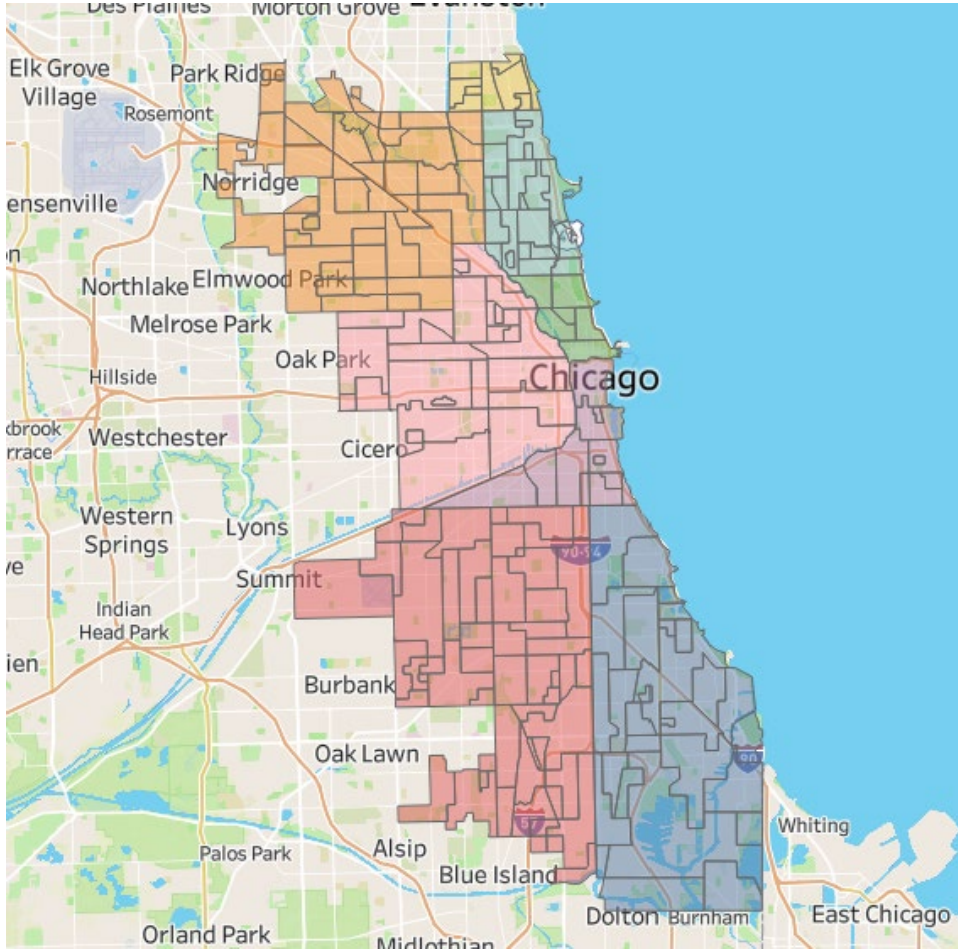


Analysis: the first iteration flagged many **Low-Price outliers** on Chicago's very heterogeneous south side.

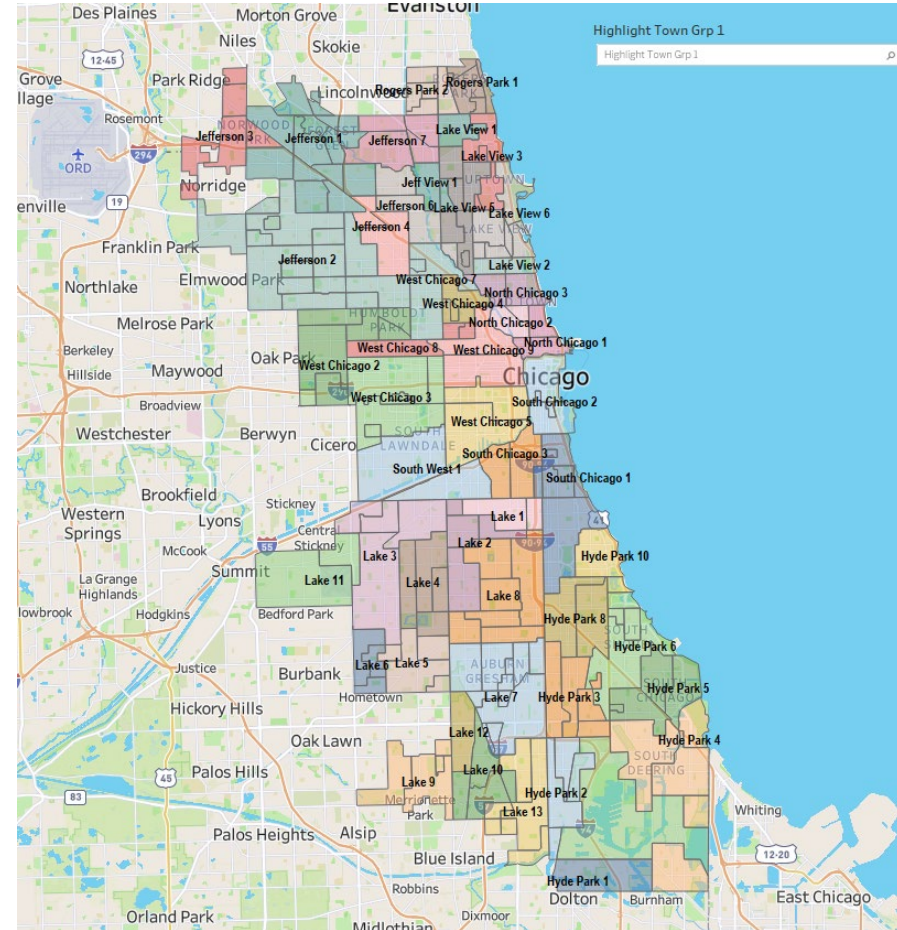




Before



After



Lesson: Test your hypotheses.



Test ideas with **domain experts**.

Domain experts can validate that your fancy statistical approaches are making the right assumptions.

Make data **visible**.

Leverage the visual system to show data in a way that helps analysts search and compare the right data.

Build on the **partnership**.

Eventually, we want domain experts to review these statistically tagged outliers.

Consult with your **data engineer** early and often.

Better efficiency and a better partnership.

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We've improved the data and modeling. **How can we earn trust after years of public mistrust and documented assessment inequities?**

Problem: Data errors can cause incorrect model estimates

DESCRIPTION	Two or more story residence, over 62 years, up to 2,200 sq.ft. 🤯
RESIDENCE TYPE	Two Story
USE	Single Family
APARTMENTS	0
EXTERIOR CONSTRUCTION	Masonry
FULL BATHS	2
HALF BATHS	0
BASEMENT ¹	Full
ATTIC	None
CENTRAL AIR	Yes
NUMBER OF FIREPLACES	0
GARAGE SIZE/TYPE ²	None
AGE	130
BUILDING SQUARE FOOTAGE	2,343 🤯

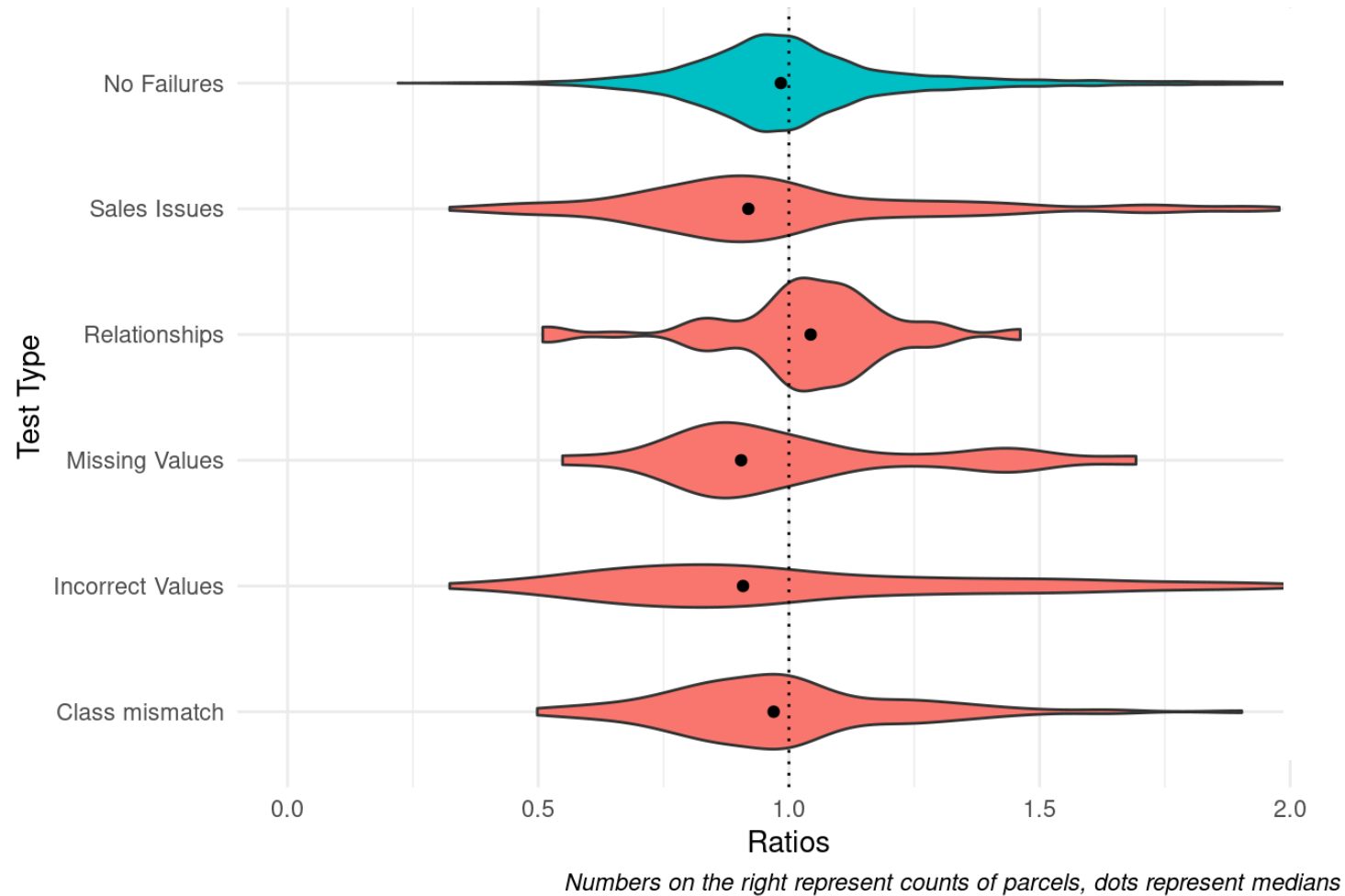


Problem: Data errors can cause incorrect model estimates



	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
5	PIN	Card	Class	Original Prediction	API Prediction	Rounded API Prediction	Town	Nbhd	Bldg. Sqft.	F. Bal		Sqft.	Fireplaces	Elementary District GEOID	Secondary District GEOID
6	16-05-100-013-0000	1	203	\$433,148			27	27010	1534	1	1930	5440	0	1729250	1729280
7	16-05-100-014-0000	1	204	\$584,156			27	27010	2341	2	1926	5440	0	1729250	1729280
8	16-05-100-015-0000	1	204	\$646,074			27	27010	3044	2	1928	5440	0	1729250	1729280
9	16-05-100-016-0000	1	204	\$525,322			27	27010	1813	2	1926	5440	0	1729250	1729280
10	16-05-100-017-0000	1	204	\$580,201			27	27010	2191	2	1926	5400	0	1729250	1729280
11	16-05-100-018-0000	1	204	\$572,309			27	27010	2236	1	1926	5400	1	1729250	1729280
12	16-05-100-019-0000	1	204	\$493,478			27	27010	1806	2	1927	5360	0	1729250	1729280
13	16-05-100-020-0000	1	204	\$556,599			27	27010	2228	1	1927	5360	0	1729250	1729280
14	16-05-100-021-0000	1	204	\$632,640			27	27010	2881	2	1926	5360	0	1729250	1729280
15	16-05-100-022-0000	1	205	\$578,059			27	27010	2038	1	1926	5320	1	1729250	1729280
16	16-05-100-023-0000	1	206	\$823,810			27	27010	3144	3	1926	5644	1	1729250	1729280
17	16-05-100-024-0000	1	204	\$692,321			27	27010	2982	2	1926	6000	0	1729250	1729280
18	16-05-100-025-0000	1	204	\$596,513			27	27010	2292	2	1931	6000	1	1729250	1729280
19	16-05-100-026-0000	1	205	\$579,055			27	27010	2167	1	1926	6000	0	1729250	1729280
20	16-05-100-027-0000	1	206	\$665,095			27	27010	2246	2	1925	6000	1	1729250	1729280
21	16-05-100-028-0000	1	205	\$489,079			27	27010	1490	1	1927	6000	1	1729250	1729280
22	16-05-100-029-0000	1	206	\$680,517			27	27010	2417	2	1926	6000	1	1729250	1729280
23	16-05-100-030-0000	1	205	\$591,883			27	27010	2088	1	1927	6000	1	1729250	1729280
24	16-05-100-034-0000	1	206	\$706,552			27	27010	2444	2	1928	6366	1	1729250	1729280
25	16-05-100-035-0000	1	205	\$635,654			27	27010	2132	1	1926	9000	1	1729250	1729280
26	16-05-100-036-0000	1	206	\$843,737			27	27010	3394	2	1926	9000	1	1729250	1729280
27	16-05-101-008-0000	1	205	\$463,038			27	27030	2052	1	1955	5000	0	1729250	1729280
28	16-05-101-009-0000	1	203	\$384,178			27	27030	1442	1	1924	5000	0	1729250	1729280
29	16-05-101-010-0000	1	204	\$517,909			27	27030	2510	2	1924	5000	1	1729250	1729280
30	16-05-101-011-0000	1	203	\$405,720			27	27030	1623	2	1924	5000	0	1729250	1729280
31	16-05-101-012-0000	1	204	\$453,918			27	27030	2110	2	1924	5000	0	1729250	1729280

Problem: Data errors are associated with worse sales ratios



Solution: Data testing

Automated tests generate spreadsheets of **data errors** for staff to review



Test description fields		Unique identifier fields						Problematic fields			
These fields identify a failing test.		These fields identify the row that is failing a test.						These fields contain values that are causing the test to fail.			
source_table	description	taxyr	parid	card	lline	class	township_code	external_calc_rcnld	external_rcnld	wbfp_o	user20
comdat	external_calc_rcnld (Net Market Value) should not be 0	2024	14281050980000	1		597	73	0			
comdat	external_rcnld (Full Market Value) should not be 0	2024	14281050980000	1		597	73		0		
dweladat	wbfp_o (Number of Fireplaces) should be between 0 and 6	2024	14204120030000	1		211	73			8	
dweladat	wbfp_o (Number of Fireplaces) should be between 0 and 6	2024	14204120040000	1		211	73			8	
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14204120310000	2		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14194240040000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14294070730000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14203170400000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14292040090000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14193120450000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14213120200000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14194250130000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14283120050000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14202070260000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14203110160000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14073070010000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14183000410000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14202070270000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14302000160000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14193110440000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14184230010000	1		212	73				0
dweladat	user20 (No. of Commercial Units) should not be 0 when class is 212	2024	14193130390000	1		212	73				0

incorrect_values

missing_values

class_mismatch_or_issue

duplicate_records

relationships

sales_issues

incorrect_values

missing_values

class_mismatch_or_issue

duplicate_records

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sales_issues



Solution: QC reporting

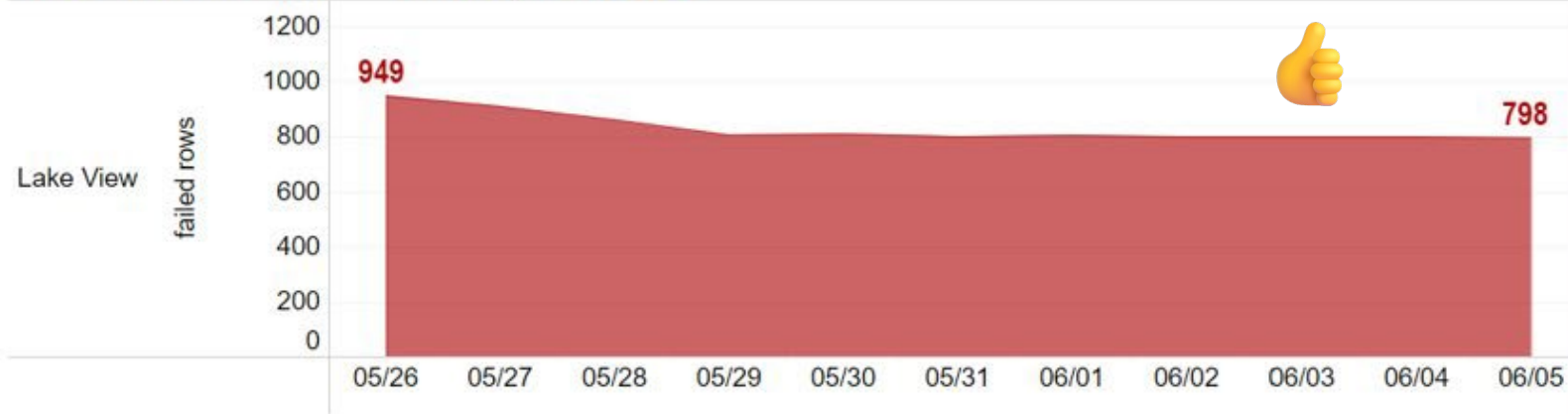
Automated scripts generate spreadsheets of suspicious data for staff to review



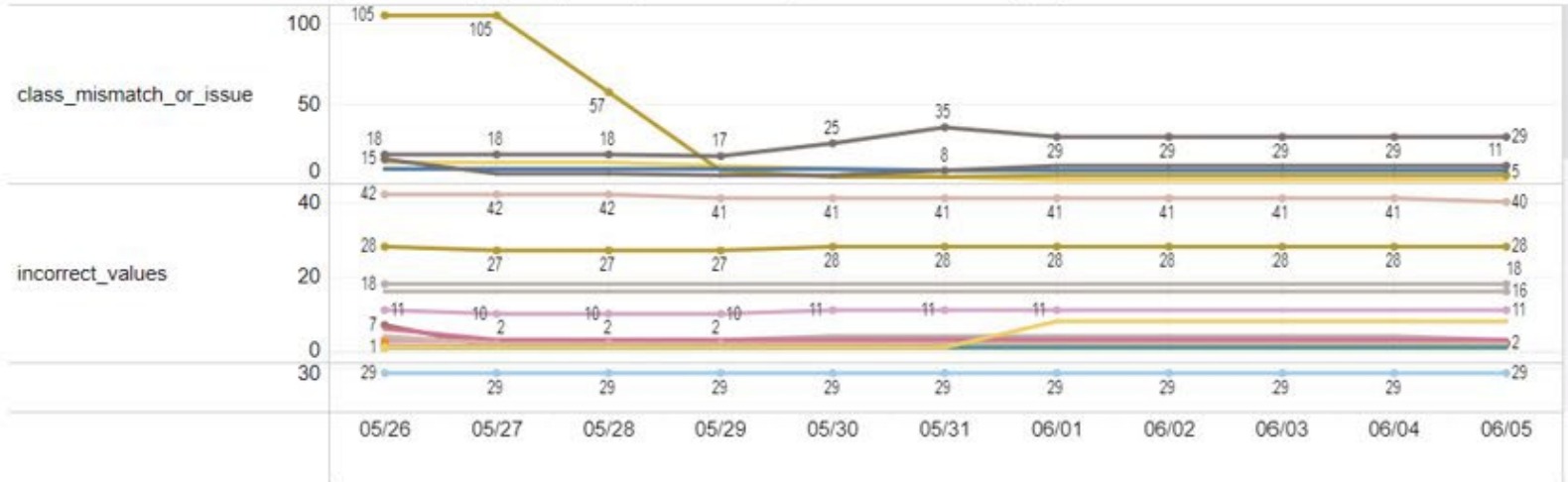
A	B	C	D	F	H	I	J	K	L	M	N	O	P	Q	R	S
PARID	TAXYR	TOWNSHIP	CLASS	Reason for Change	2023 LMV	2023 BMV	2023 Total MV	2023 LAV	2023 BAV	2023 Total AV	2024 LMV	2024 BMV	2024 Total MV	2024 LAV	2024 BAV	2024 Total AV
17032280354032	2024	74	299		0	20000	20000	0	2000	2000	0	210	210	0	21	21
17104010051754	2024	76	599		2815	111224	114039	704	27806	28510	1	111224	111225	0	27806	27806
17101350391002	2024	74	599		651	101928	102579	163	25482	25645	1	219724	219725	0	54931	54931
17101350391335	2024	74	599		693	108424	109117	173	27106	27279	1	233736	233737	0	58434	58434
17032280344035	2024	74	299		0	2517290	2517290	0	251729	251729	0	607690	607690	0	60769	60769
17101120111385	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17032280344006	2024	74	299		0	714220	714220	0	71422	71422	0	172420	172420	0	17242	17242
17101120111477	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17101120111433	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17101120111466	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17101120111547	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17032280354049	2024	74	299		0	20000	20000	0	2000	2000	0	210	210	0	21	21
17032280344013	2024	74	299		0	1325040	1325040	0	132504	132504	0	319870	319870	0	31987	31987
17101120111417	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17101120111578	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17101120111371	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
25203060010000	2024	72	202		0	0	0	0	0	0	0	176920	176920	0	17692	17692
17101120111491	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17032280354031	2024	74	299		0	20000	20000	0	2000	2000	0	210	210	0	21	21
17032280354019	2024	74	299		0	20000	20000	0	2000	2000	0	210	210	0	21	21
17101120111298	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17101120111243	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17101120111233	2024	74	299		10	20	30	1	2	3	0	20	20	0	2	2
17103080240000	2024	76	797	30	1	4	5	0	1	1	1	10	11	0	1	1
17032210121177	2024	74	599		1780	5064	6844	445	1266	1711	1	10860	10861	0	2715	2715
17211200470000	2024	77	100		270	0	270	27	0	27	0	0	0	0	0	0

Solution: Tracking data test failures over time

Total Failures in Selected Township: Lake View



Number of rows failing QC tests, grouped by Township then test category



Lesson: Data departments can make QC more effective



Build on existing processes

Valuations staff check for mistakes before townships get mailed, so we send them our reports then

Meet stakeholders where they are

Valuations staff are comfortable with spreadsheets, so we output spreadsheets for them to review

Automate the boring stuff

We generate our spreadsheets using Python scripts, so staff don't have to waste time running queries

Test early and often

"There couldn't possibly be any negative assessed values in our database, right...?"

Data and Modeling

Outlier sales exist. How should we identify and exclude them?

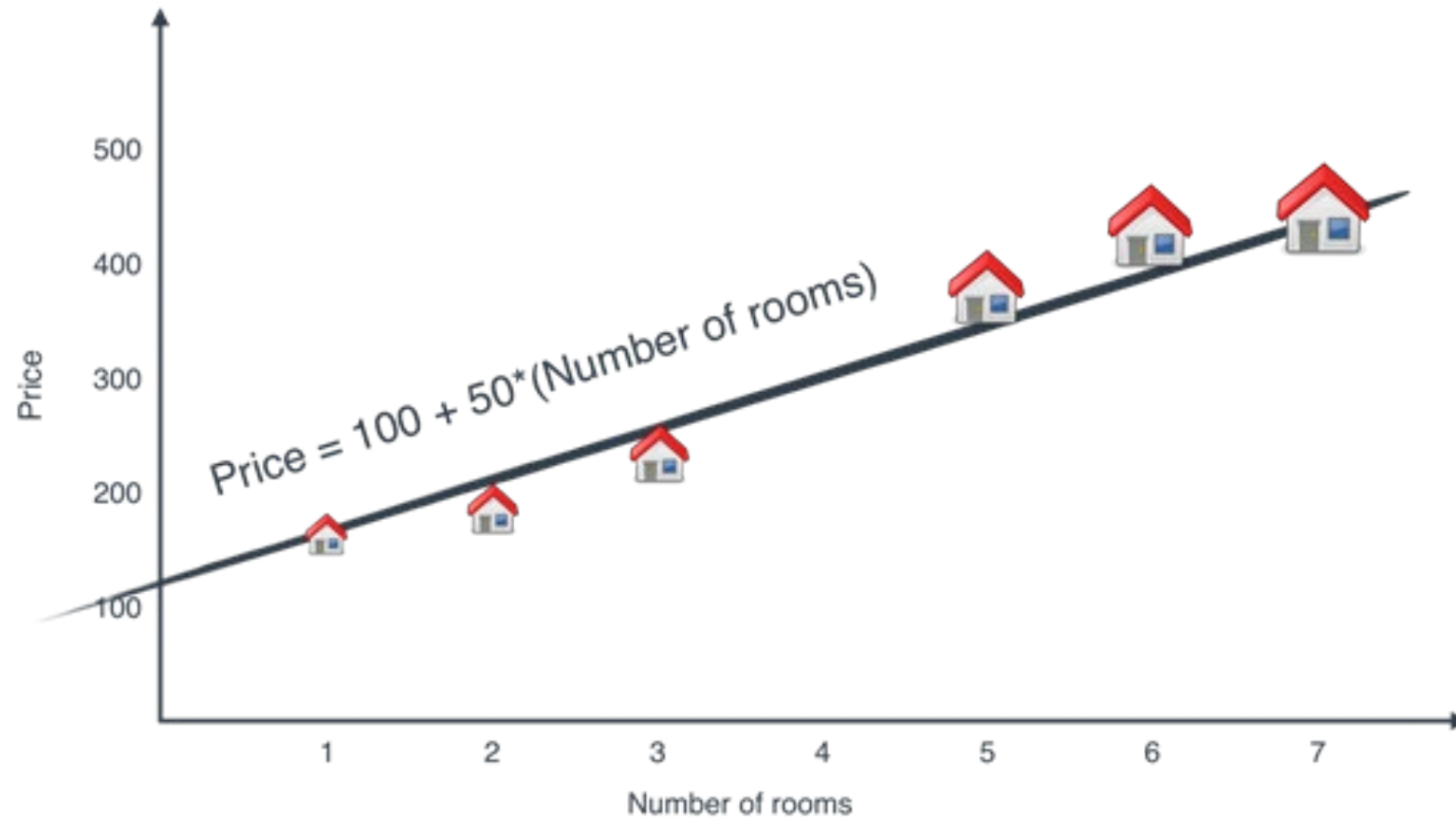
Data errors exist. How do we find them?

Past ratio stats have room for improvement. Does a **non-linear model** help?

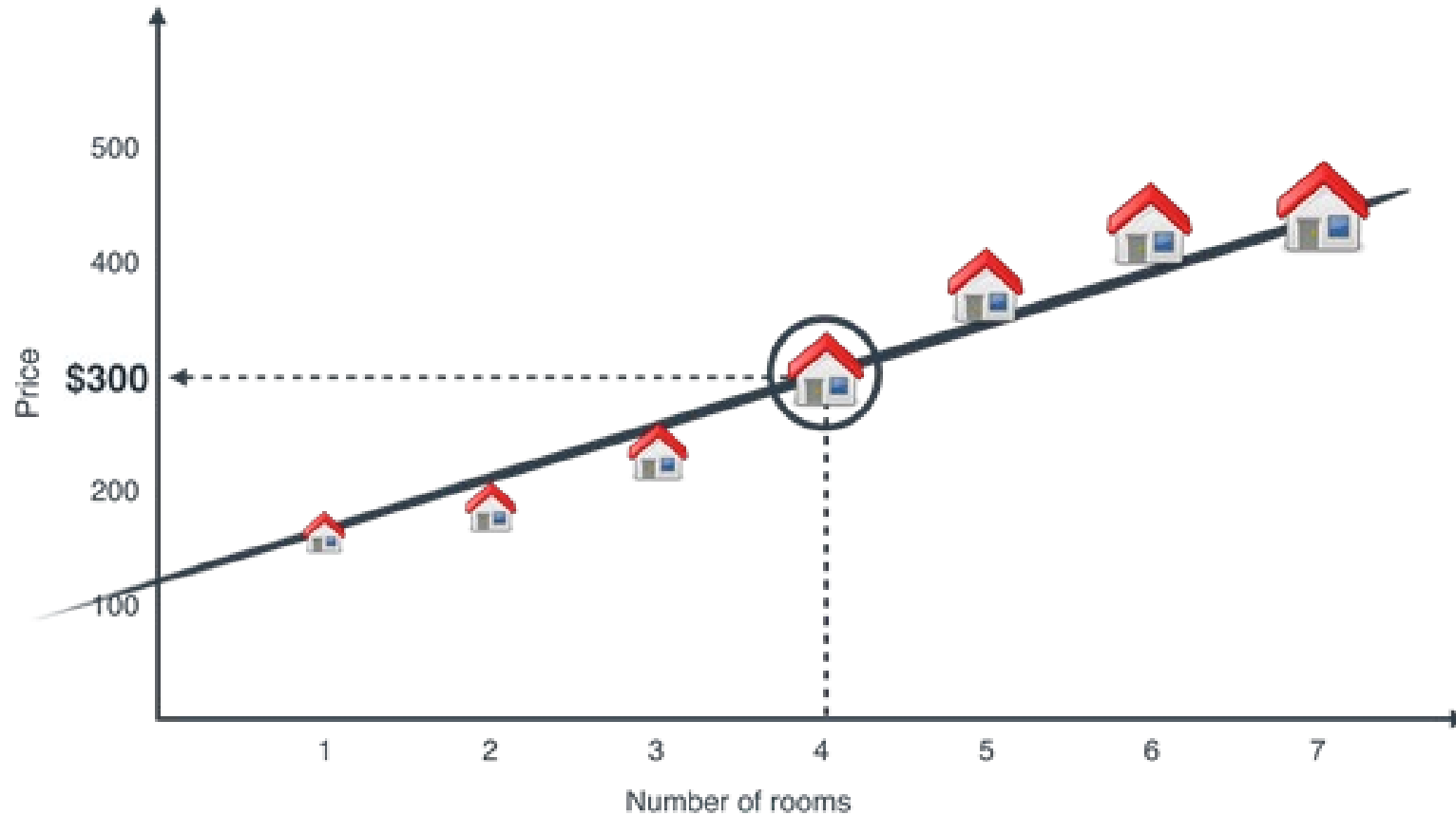
Transparency problems

We've improved the data and modeling. **How can we earn trust after years of public mistrust and documented assessment inequities?**

Linear Model: Training



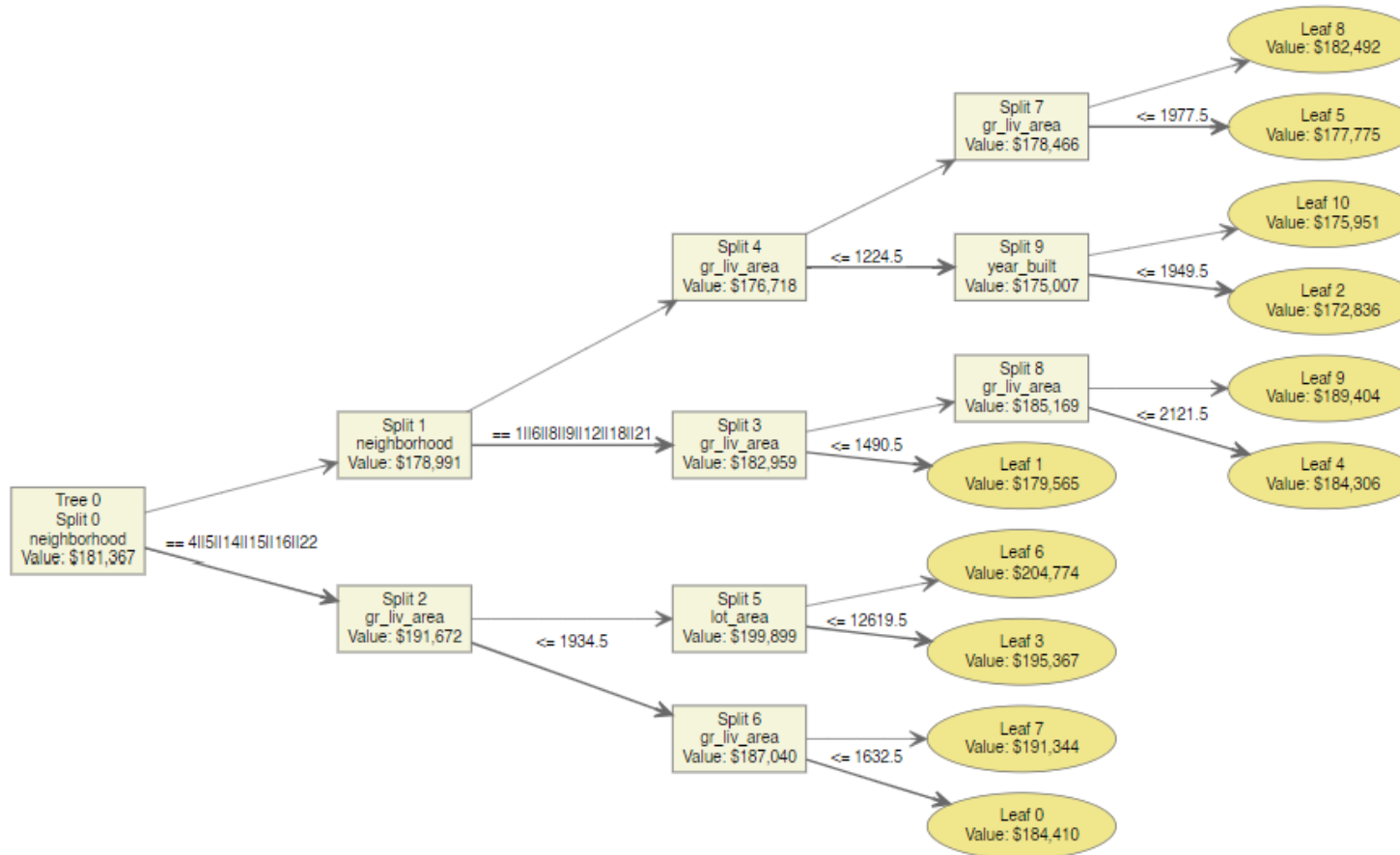
Linear Model: Prediction



Pros: Highly interpretable and generalizable.

Cons: sometimes not as performant, and conditional averaging may introduce inequities

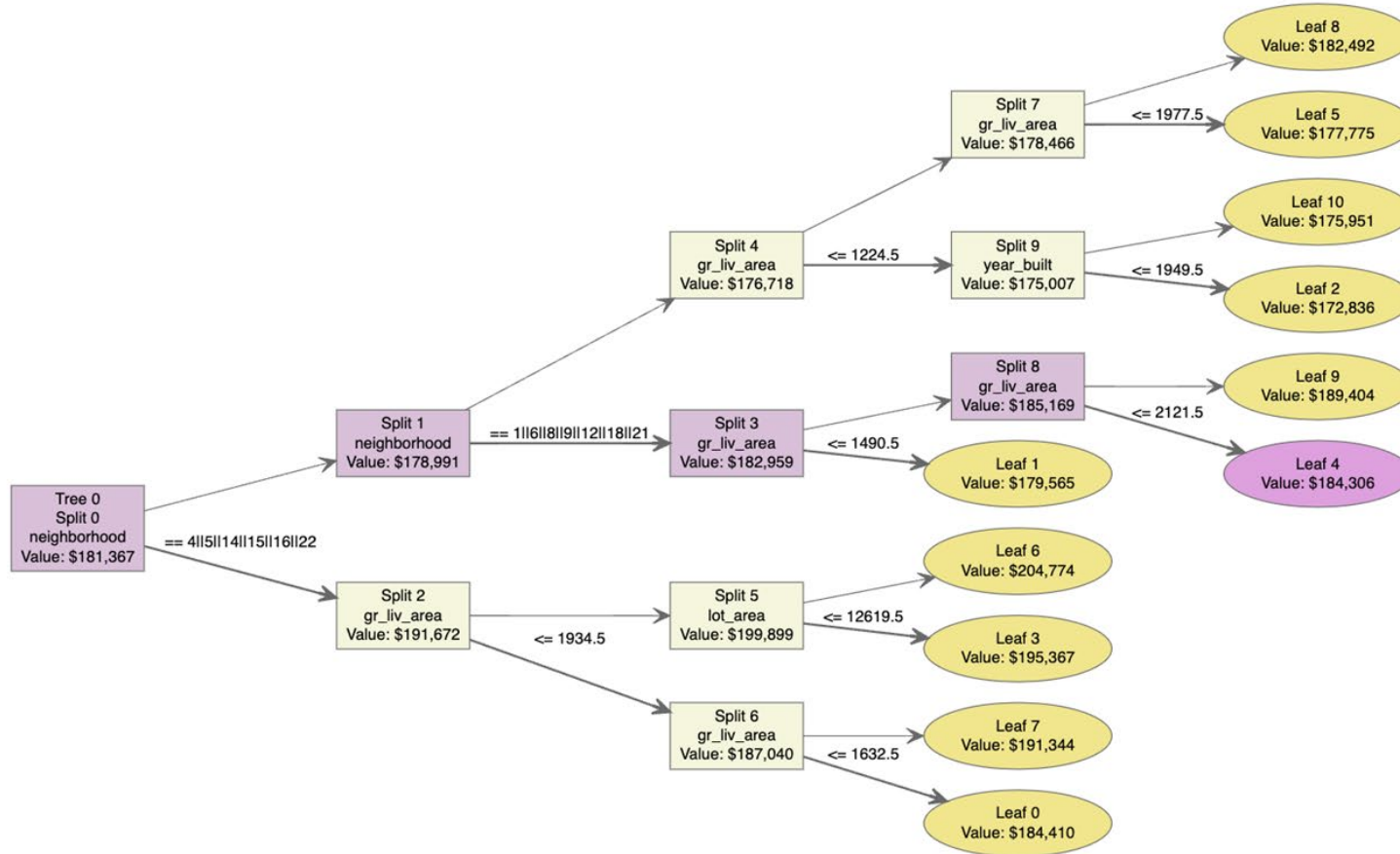
Non-Linear Model: Training



Decision tree models make **decisions** about where to **split** the data into two partitions using “if-then” rules.

Partitions are split into two again with more “if-then” rules until more partitions don’t add much information.

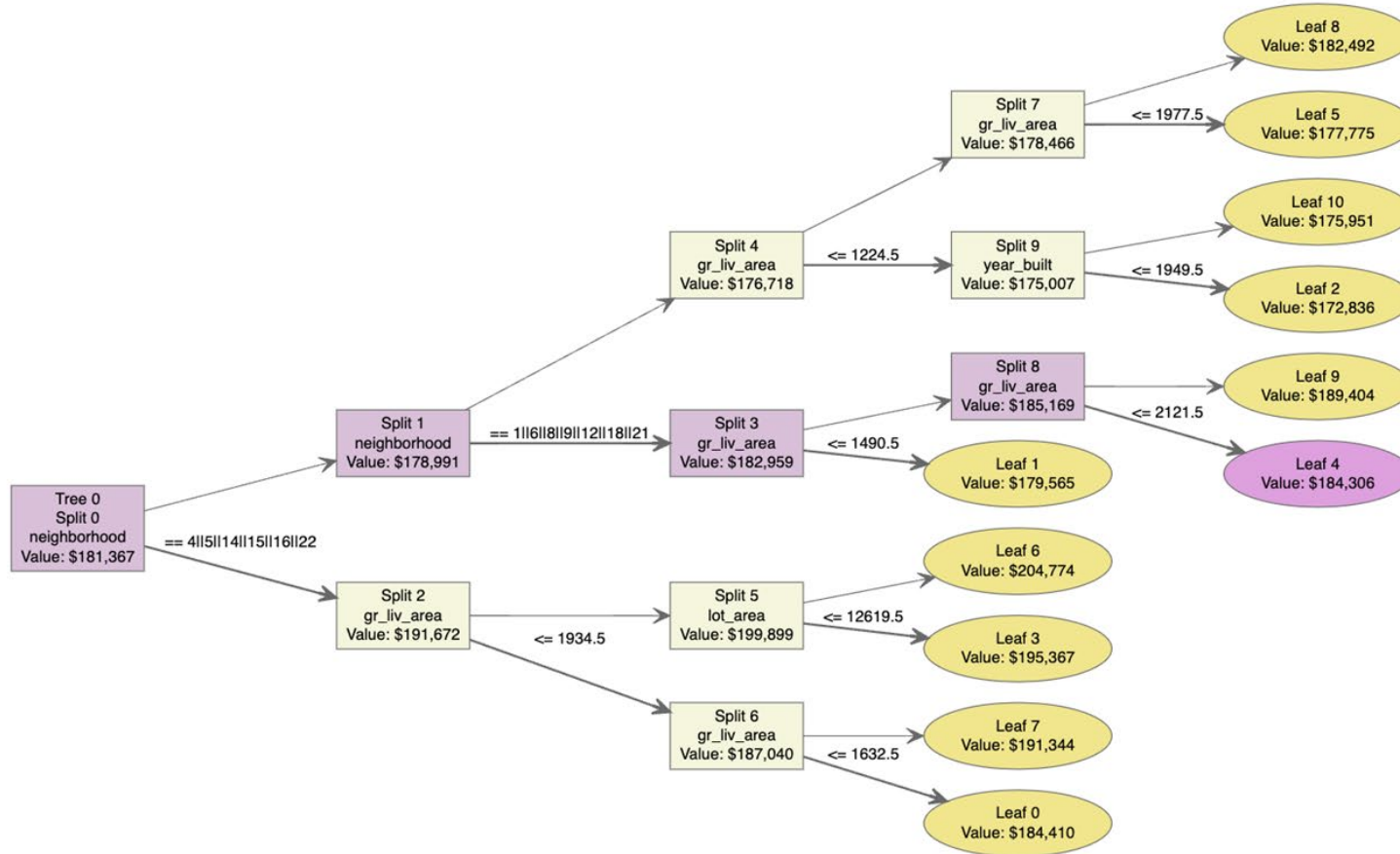
Tree-Based Model: Prediction



Pros: Advantageous for non-linear relationships, handles categorical variables well, can outperform linear models.

Cons: requires careful tuning to avoid over-fitting.

Tree-Based Model: Prediction



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Comparable sales ?

Data and Modeling

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How do we earn trust?

Outlier sales

- Analytic approaches
- Technical implementation

Data errors

- Our process to automatically tag and fix data errors

A new model

- Linear vs. tree-based model: does it matter?

Explainability

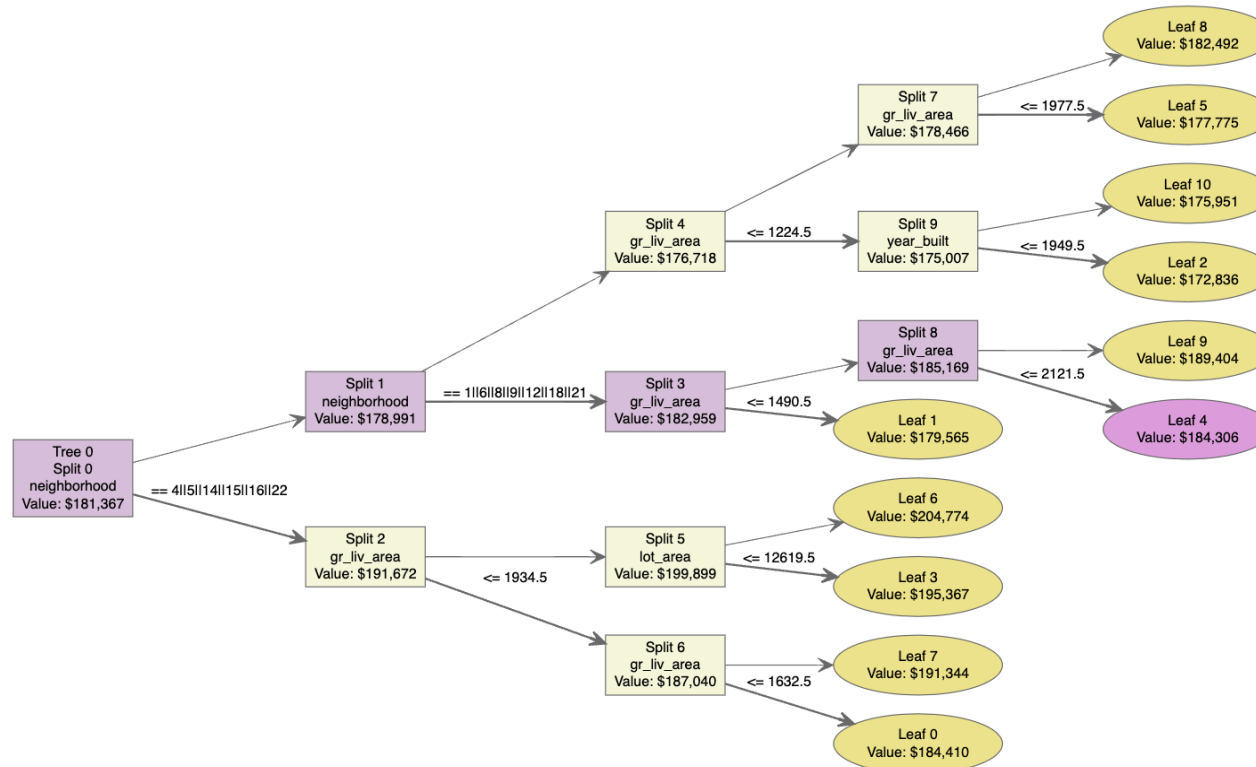
- A new sale comps algorithm

“What sales did the model use to assess my home?”

Explainable AI (XAI): AI should be able to produce details that make its functions easy to understand for its intended audience.

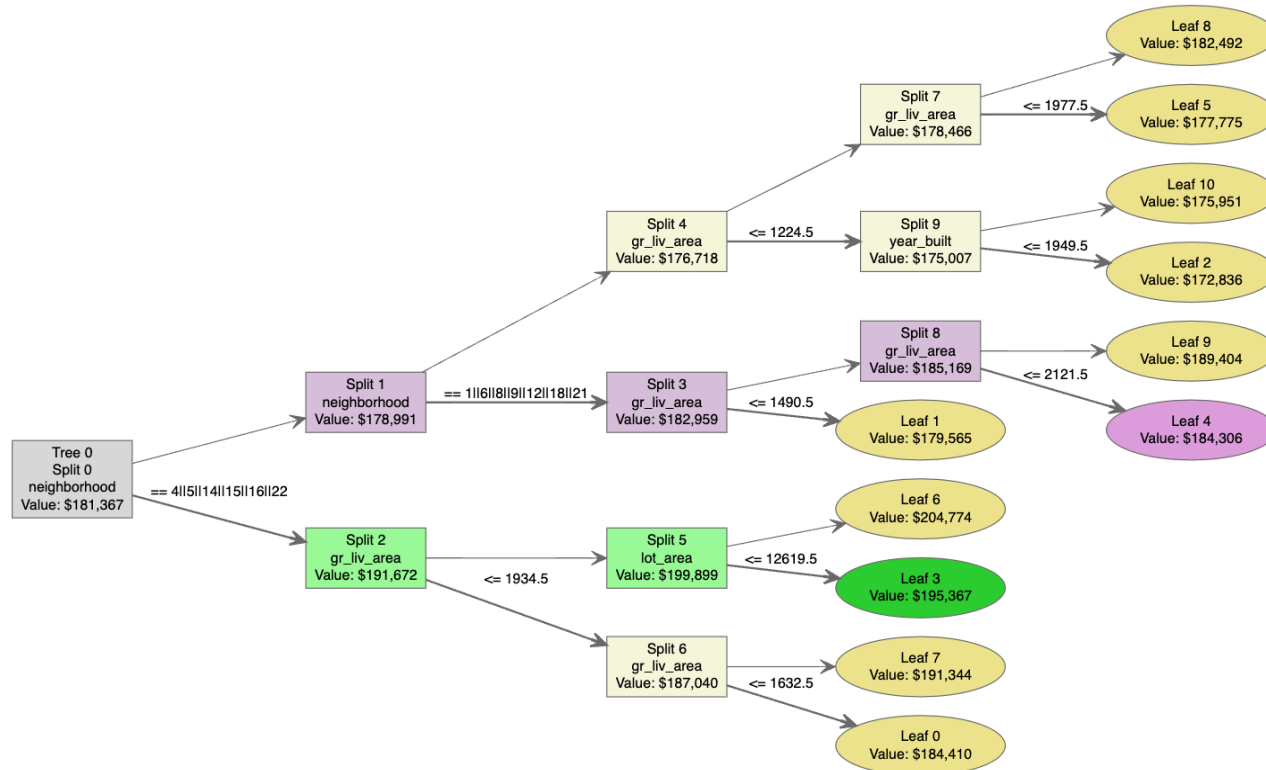
We created an algorithm to analyze our model's decision trees...

ID	Livable Area	Lot Area	Year Built	Condition	Neighborhood	Pred. Value
2	1,629	13,830	1997	4	6	\$184,306



We created an algorithm to analyze our model's decision trees...

ID	Livable Area	Lot Area	Year Built	Condition	Neighborhood	Pred. Value
2	1,629	13,830	1997	4	6	\$184,306
75	1,968	12,003	2009	4	14	\$195,367



... to find sales that each decision tree thinks are similar...



ID	Livable Area	Lot Area	Year Built	Condition	Neighborhood	Pred. Value
3	1,604	9,978	1998	5	6	\$184,306
14	1,960	7,851	2002	4	6	\$184,306
21	2,110	8,880	1994	4	9	\$184,306
48	1,675	15,263	1959	4	18	\$184,306
55	1,694	10,475	2008	4	1	\$184,306
60	1,978	10,389	2003	4	1	\$184,306
61	2,098	9,375	1997	4	1	\$184,306
62	1,661	12,137	1998	4	1	\$184,306
70	1,652	19,645	1994	5	12	\$184,306
84	1,571	7,837	1993	6	6	\$184,306

... then make comparisons *between* decision trees...



ID	Tree 0	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Tree 6	Tree 7	Tree 8	Tree 9
2	\$184,306	\$3,322	\$2,791	\$1,779	\$1,853	\$1,928	\$1,398	\$1,483	-\$450	\$605
ID	Tree 0	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Tree 6	Tree 7	Tree 8	Tree 9
3	\$184,306	\$3,322	\$2,791	\$1,779	\$1,853	\$1,928	-\$1,645	\$1,483	-\$450	\$605
14	\$184,306	\$3,322	\$9,008	\$1,779	\$4,435	\$7,398	-\$1,645	\$1,483	\$1,928	\$1,288
21	\$184,306	\$3,322	\$9,008	\$1,779	\$4,435	\$7,398	\$2,164	\$6,202	\$1,928	\$605
48	\$184,306	\$3,322	-\$2,739	\$1,779	\$1,853	-\$2,057	\$2,164	-\$1,975	\$3,036	\$605
55	\$184,306	\$10,207	\$7,507	\$1,779	\$1,853	\$1,928	\$2,164	\$5,202	\$5,038	\$3,666
60	\$184,306	\$3,322	\$9,008	\$1,779	\$4,435	\$7,398	\$2,164	\$1,483	\$3,036	\$3,666
61	\$184,306	\$3,322	\$9,008	\$1,779	\$4,435	\$7,398	\$2,164	\$6,202	\$3,036	\$605
62	\$184,306	\$3,322	\$2,791	\$1,779	\$1,853	\$1,928	\$2,164	\$1,483	\$3,036	\$605
70	\$184,306	\$3,322	\$2,791	\$1,779	\$1,853	\$5,629	\$2,164	\$1,483	\$3,036	\$605
84	\$184,306	-\$1,044	\$2,791	\$1,779	\$1,853	\$1,928	-\$1,645	\$1,483	-\$450	\$605

... then make comparisons *between* decision trees...



ID	Tree 0	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Tree 6	Tree 7	Tree 8	Tree 9	Number	Percent
2	4	4	4	2	4	7	6	4	6	4		

ID	Tree 0	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Tree 6	Tree 7	Tree 8	Tree 9	Number	Percent
3	T	T	T	T	T	T	F	F	F	F	6 / 10	60%
14	T	T	F	F	F	F	F	F	F	F	2 / 10	20%
21	T	T	F	F	F	F	F	F	F	F	2 / 10	20%
48	T	T	F	F	F	F	F	F	F	F	2 / 10	20%
55	T	F	F	F	F	F	F	F	F	F	1 / 10	10%
60	T	T	F	F	F	F	F	F	F	F	2 / 10	20%
61	T	T	F	F	F	F	F	F	F	F	2 / 10	20%
62	T	T	T	T	T	T	F	F	F	F	6 / 10	60%
70	T	T	T	T	T	F	F	F	F	F	5 / 10	50%
84	T	F	T	T	T	T	F	F	F	F	6 / 10	60%

... then make comparisons *between* decision trees...



	Tree 0	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Tree 6	Tree 7	Tree 8	Tree 9
Weight	71.2%	4.7%	4.2%	3.8%	3.4%	3.1%	2.8%	2.5%	2.2%	2.0%

ID	Tree 0	Tree 1	Tree 2	Tree 3	Tree 4	Tree 5	Tree 6	Tree 7	Tree 8	Tree 9	Sim. Score
3	71.2%	4.7%	4.2%	3.8%	3.4%	3.1%	0.0%	0.0%	0.0%	0.0%	90.46%
14	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
21	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
48	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
55	71.2%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	71.18%
60	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
61	71.2%	4.7%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	0.0%	75.88%
62	71.2%	4.7%	4.2%	3.8%	3.4%	3.1%	0.0%	0.0%	0.0%	0.0%	90.46%
70	71.2%	4.7%	4.2%	3.8%	3.4%	0.0%	0.0%	0.0%	0.0%	0.0%	87.37%
84	71.2%	0.0%	4.2%	3.8%	3.4%	3.1%	0.0%	0.0%	0.0%	0.0%	85.76%

... and tell us which sales the model thinks are comparable



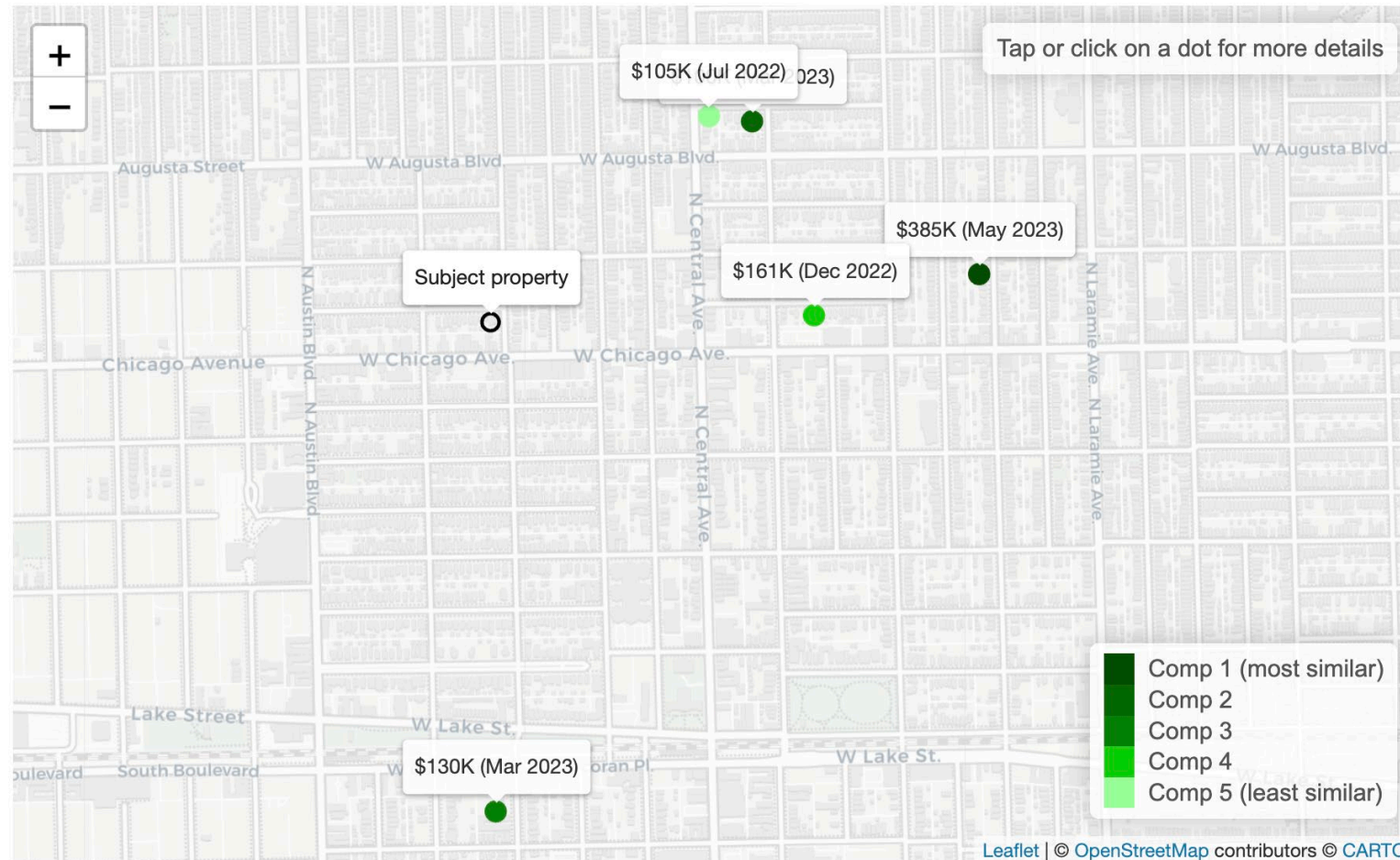
ID	Livable Area	Lot Area	Year Built	Condition	Neighborhood	Sale Price	Sim. Score
2	1,629	13,830	1997	4	6	\$189,900	

ID	Livable Area	Lot Area	Year Built	Condition	Neighborhood	Sale Price	Sim. Score
3	1,604	9,978	1998	5	6	\$195,500	90.46%
62	1,661	12,137	1998	4	1	\$224,900	90.46%
70	1,652	19,645	1994	5	12	\$203,135	87.37%
84	1,571	7,837	1993	6	6	\$178,000	85.76%
14	1,960	7,851	2002	4	6	\$216,500	75.88%
21	2,110	8,880	1994	4	9	\$205,000	75.88%
48	1,675	15,263	1959	4	18	\$173,000	75.88%
60	1,978	10,389	2003	4	1	\$318,000	75.88%
61	2,098	9,375	1997	4	1	\$240,000	75.88%
55	1,694	10,475	2008	4	1	\$245,350	71.18%

These comps show us *how the model predicted value for a property*



Top 5 comparable homes



The average sale price of the top 5 comparable homes was **\$176,200** at **\$88/sq.ft.**

Based on these and other sales, the model that ran on March 17, 2024 predicted that the value of this home as of lien date January 1, 2024 should be **\$192,072** at **\$96/sq.ft.**

These comps show us *how the model predicted value for a property*



Live Demo

Measuring Impact: New Statistical Methods for Assessment Accuracy and Transparency



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github.com/ccao-data

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